

SIMATS ENGINEERING ASSESSMENT TOOL

COURSE NAME: Theory of Computation

COURSE CODE: CSA13

COL - A52

COURSE OUTCOME COVERED

CO1: Analyze fundamental mathematical concepts of automata theory for formal languages and decision problems(BL:3)

Assessment Tool Weightage: 20%

QUESTIONS: 5
Duration : 1 Hour

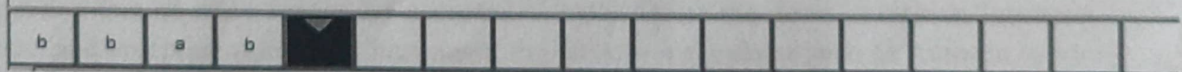
Total Marks:25

Simulation

Code of Conduct:

I Om Sai Reddy / 192372252 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

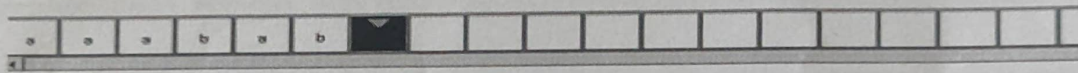
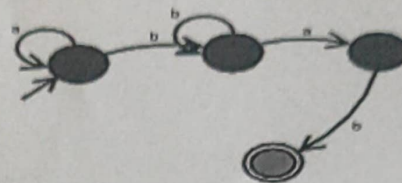
Signature of the student: Sai Reddy



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3. Design DFA using simulator to accept the string the end with ab over set $\{a,b\}$ $W = aabab$

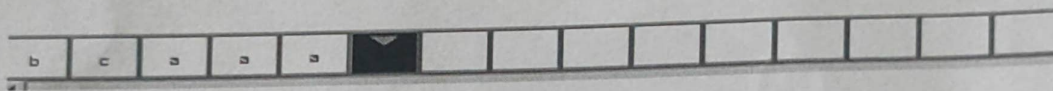
aabab



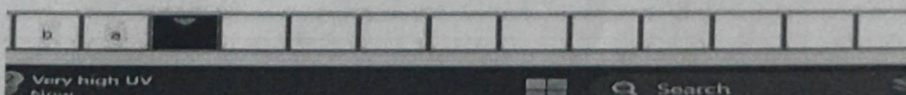
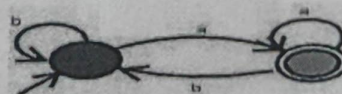
4. Design a Deterministic Finite Automaton (DFA) using a simulator to accept only the input strings "c", "bc", and "bcaaa" over the alphabet $\{a, b, c\}$. The automaton should begin from an initial state and transition through intermediate states based on the sequence of input symbols, ensuring that each valid string follows a unique path. For example, the string "c" should be accepted directly from the start state, while "bc" should pass through states representing "b" and then "c". Similarly, "bcaaa" should be processed through a sequence of states that track each additional "a" after "bc". Any deviation from these exact patterns should lead to a dead state, ensuring that no other strings are accepted. Clearly define the states, transition functions, start state, and accepting states, and implement the DFA in a simulator such as Autosim to verify its correctness.

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bcccc



5. Design a Deterministic Finite Automaton (DFA) using a simulator to accept all strings over the alphabet $\{a, b\}$ that begin with either "a" or "b". The automaton should start in an initial state and immediately transition to an accepting state upon reading the first input symbol, since any valid string must start with either "a" or "b". After the first symbol is processed, the DFA should remain in an accepting state regardless of the subsequent sequence of "a" and "b", thereby allowing any combination of characters to follow. Additionally, include a dead state to handle invalid or empty inputs if required by the simulator. Clearly define the states, transition rules, start state, and accepting states, and verify the design using a simulator such as Autosim to ensure that all valid strings are accepted and invalid ones are rejected



91/100

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